OATM: Occlusion-Adaptive Temporal Memory: Camera-Only Track Continuity Under Visual Occlusion

Noor Basfar, Ahmed Bahaj, Tanveer Hussain

King Abdullah University of Science and Technology · Saudi Research Science Institute

The Problem

Single-frame detection cannot distinguish temporary occlusion from departure. **When the object reappears, the tracker may assign a new identity**. This matters because nearly 40% of nuScenes instances are below 80% visibility. These identity breaks weaken reliable trajectory continuity in crowded driving scenes.

Experimental setup

Shared input: frozen YOLO detections from 10 CAM_FRONT scenes (49,436 detections)
Identical detections: OATM, ByteTrack-5, and ByteTrack-12
Online OATM uses camera detections and track history only.
LiDAR was used only for offline evaluation, never as an input to the tracker.

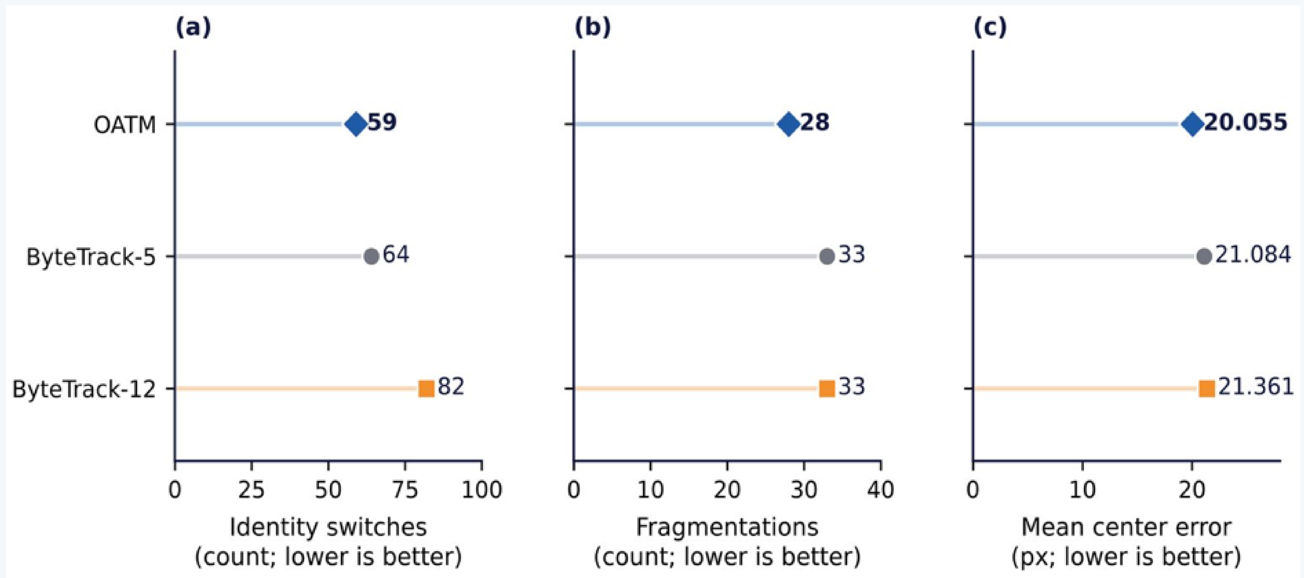


Figure 2. Identity switches, fragmentations and mean center error. OATM is best on all three.

Methodology

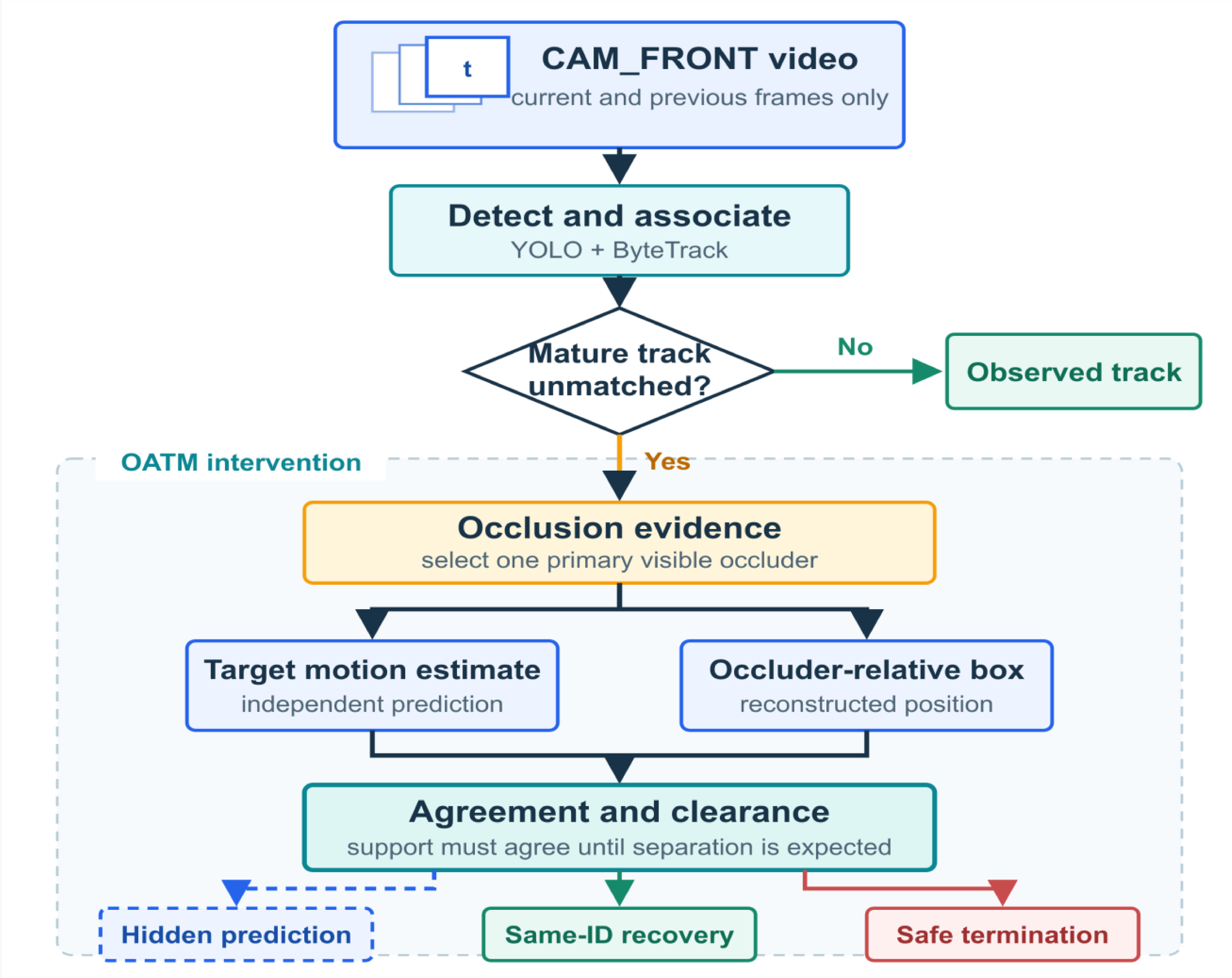


Figure 1. Occlusion evidence and dual prediction govern recovery or termination.

Results

Method · IoU 0.30	Prec.	Rec.	F1	MOTA	IDF1
OATM	**84.1%**	**26.6%**	**40.4%**	**18.4%**	**30.3%**
ByteTrack-5	73.4%	27.6%	40.1%	14.2%	29.6%
ByteTrack-12	55.3%	28.6%	37.7%	1.1%	27.4%

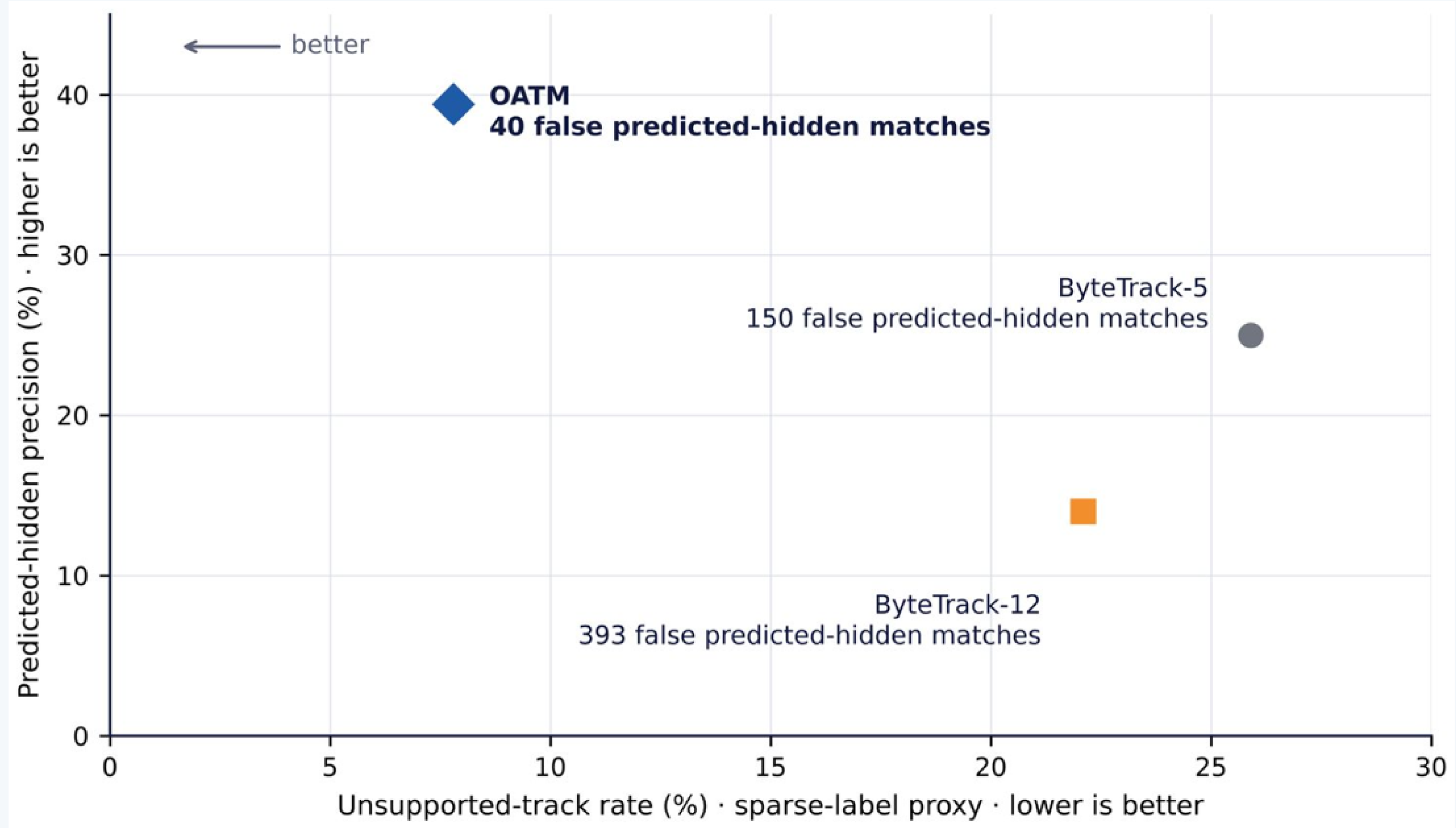


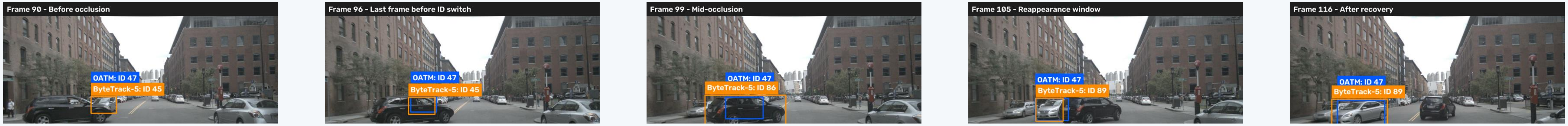
Figure 3. Selectivity of predicted-hidden persistence: fewest unsupported tracks.

Conclusion

Dual prediction and a visible occluder explain temporary disappearance causally. **Same-ID recovery and bounded termination keep persistence conservative**. Identity is therefore preserved only while the occlusion explanation remains supported.

Case Study: One Real Occlusion Event

ByteTrack-5 switches identity three times (45 → 57 → 86 → 89) while **OATM holds ID 47 throughout.**



Discussion

- OATM's precision and identity gains reflect selective persistence, not longer retention.
- For this continuity task, camera-only tracking offers a lower-cost alternative; **LiDAR is used only for offline evaluation**.
- Severe-visibility recall and the 10-scene scale limit generalization; full nuScenes evaluation is next.

References

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